

14.9 Solving One-Step Inequalities Using Addition and Subtraction

Lesson Introduction

 Benchmark	MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.	 Learning Targets	• I can relate the solution set of an inequality to its graph on a number line. • I can relate the solution set of an inequality to the context it represents. • I can determine the minimum or maximum value that makes an inequality true. • I can solve one-step inequalities in one variable using addition or subtraction. • I can relate properties of inequality to solving one-step inequalities algebraically.	
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line. MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.		MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.	
 Essential Questions	• What is the minimum or maximum value that makes the inequality true? • How do you relate the solution set of an inequality to the context it represents? • How do you determine if solutions are reasonable for the context? • How can I use properties of inequalities to solve one-step addition and subtraction inequalities? • How do I show the solution of an inequality on a number line?			
 Lesson Narrative	In this lesson, students read, graph, and interpret inequalities in the form $x + p < q$. This lesson is an exploration with scaffolding where students connect algebraic representations of an inequality to a number line and a real-world context. Students also explore properties of inequality to get an understanding of the process of solving inequalities. Then students use properties of inequality to solve one-step addition and subtraction inequalities. Students also graph the solution sets and test values to prove their solution is true.			
 Component Summary	14.9.1 Warm-Up (~5 min.)	Identify and justify which inequality does not belong (SE p. 384)	14.9.4 Exploration (10 min.)	Explore properties of inequalities (SE p. 387)
	14.9.2 Exploration (10-15 min.)	Represent inequalities in the context of real-world problems (SE p. 384)	14.9.5 Guided Instruction (10 min.)	Learn how to solve and graph one-step addition and subtraction inequalities (SE p. 388)
	14.9.3 Bring It Together (5-10 min.)	Consider constraints on the solution sets based on the context of inequalities (SE p. 386)	14.9.6 Wrap-Up (~5 min.)	Reflect on learning targets by completing a Know, Wonder, and Learned table (SE p. 390)

 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> • Solution set • Constraint • Addition property of inequality • Subtraction property of inequality • Asymmetric property of inequality <u>Additional Terms:</u> • Maximum • Minimum • Inequality		• (Optional) Number line templates • (Optional) Colored pencils • (Optional) Chart paper • (Optional) Markers	• Consider preparing sentence stems for constructive response problems.
 Teacher Tips	Common Misconceptions			Things to Consider
	• Students may struggle understanding that $x > a$ and $a < x$ are equivalent, because they are not accustomed to the variable being on the right side. • Students may struggle determining whether to shade the circle on the number line. • Students may struggle finding the maximum or minimum. • Students may struggle to understand that the value (p) for a strict inequality ($x > p$ or $x < p$) is not the minimum or maximum. • Students may have a hard time interpreting the solution sets within the context.			• Consider creating and displaying an anchor chart and word wall for students to reference throughout the lesson. Include instructions, definitions, examples, and visual models. • Consider having students practice reading the inequalities aloud from right to left and left to right.
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
 Differentiate Instruction	Think-Pair-Share In 14.9.2 Exploration, have students initially complete question 2 independently. Then have students discuss their solutions and reasonings with their partners. These discussions will help students build confidence and understanding.		MA.K12.MTR.7.1: Real-World Contexts In 14.9.3 Bring It Together, students solve, graph, and interpret inequalities based on real-world situations. Students learn that sometimes solutions do not make sense in the context of a problem. For example, if determining the possible height of a person, negative solutions do not make sense.	
	Visual, auditory, and kinesthetic entry points are built in throughout this lesson. Number lines can be physical or digital manipulatives that benefit students who learn best with hands-on resources.			
Provide English Language Learners with anchors of support, such as vocabulary cards of the words used in this lesson: minimum, maximum, inequality, solution set, constraint.				
Lesson Notes:				

14.9.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students are given four different inequalities and asked to determine which inequality does not belong and explain why. Having students raise their hands when they have a solution can discourage other students from attempting the problem. To encourage all students to participate, ask students to hold their hands over their hearts and lift a finger to signify how many solutions they have determined.



14.9.1 Warm-Up: Which One Doesn't Belong?

Four inequalities are shown.

Inequality A	Inequality B	Inequality C	Inequality D
$x \geq 5$	$12 \leq 7$	$2 > x$	$x > 1$

- Javier says that Inequality B does not belong with the others. What is one reason Javier might have said this?
The inequality B is wrong because 12 is greater than 7.



Students may need a refresher on how to read inequalities. Model how to read inequalities from left to right and from right to left. Then have students practice reading the inequalities themselves.



After initially identifying one inequality that does not belong, challenge students to think of as many reasons as they can for each inequality not belonging.

- Select a different inequality and explain why it doesn't belong.
Answers will vary.
 $2 > x$
I chose this one because in all the other equations x is greater.

14.9.2 Exploration: Representing Inequalities in Context

Component Overview: Students explore how to represent inequalities in context. Students initially match height restrictions of attractions with number lines. Next, they use substitution to determine whether values are contained in the solution sets of inequalities. Finally, students write, graph, and determine the minimum solution of an inequality that represents the height restrictions of a ride.

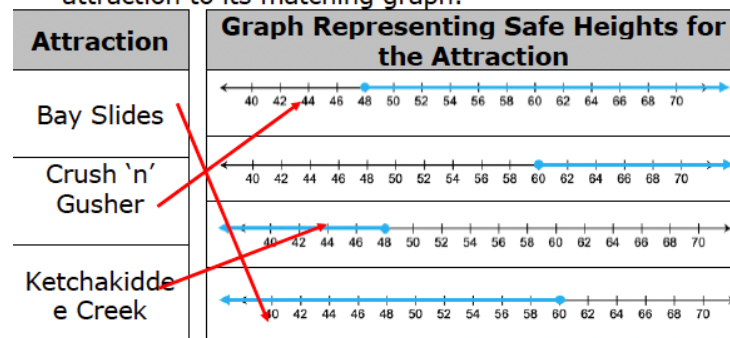


14.9.2 Exploration: Representing Inequalities in Context

Height restrictions are often in place for roller coasters and rides at amusement parks to keep guests safe. At Disney's Typhoon Lagoon Water Park, the following height restrictions are in place.

Attraction	Height Restriction
Bay Slides	Guests must be 60 inches or shorter
Crush 'n' Gusher	Guests must be 48 inches or taller
Ketchakiddee Creek	Guests must be 48 inches or shorter

1. Work with your partner to determine which graph represents the height restrictions of each attraction at Typhoon Lagoon. Then, draw an arrow from each attraction to its matching graph.



2. Nick is happy to realize that he is tall enough to go on the Crush 'n' Gusher raft ride. Nick's friend Matt is 2 inches shorter than Nick. Explain whether Matt is tall enough to go on the ride with Nick.
Answers will vary. If Nick is 48 inches tall then Matt would not be tall enough because he would be 46 inches. However, if Nick is 50 inches or taller then Matt is tall enough because he would be at least 48 inches.



- How do you determine from the height restriction chart whether the circle is closed or open?
- How do you determine from the height restriction chart which direction to shade?
- What values make sense in this context?



Think-Pair-Share

Have students initially complete question 2 independently. Then have students discuss their solutions and reasonings with their partners. These discussions will help students build confidence and understanding.

14.9.2 Exploration: Representing Inequalities in Context (continued)

3. Complete the statements.

If Nick is n inches tall, then Matt is ☐ $n + 2$
☐ $n - 2$ inches tall.

For Matt to be tall enough to go on Crush 'n' Gusher,

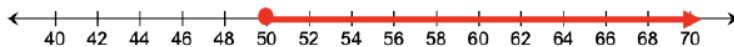
the value of ☐ $n + 2$
☐ $n - 2$ must be ☐ less than
☐ greater than or

equal to ☐ 48.
☐ 60. Therefore, the solutions to the

inequality ☐ $n + 2 \leq$
☐ $n + 2 \geq$
☐ $n - 2 \leq$
☐ $n - 2 \geq$ ☐ 48
☐ 60 represent Nick's height

that mean Matt could also ride the Crush 'n' Gusher.

4. On the number line, show all the possible heights that Nick could be if Matt is also tall enough to go on the ride.



5. What is the minimum height Nick must be in order for Matt to be able to go on the ride as well?
Nick must be at least 50 inches for Matt to go on the ride as well.

Julia noticed the graph also represents the solutions to the inequality $n \geq 50$.

6. Work with your partner to explain why the solutions to the inequality $n - 2 \geq 48$ can be represented the same way on a number line as $n \geq 50$.
Answers will vary. The solution to $n - 2 \geq 48$ is the same set of numbers as the solution to $n \geq 50$, therefore the graphs will be the same.



For question 3, consider modeling Guess and Test for students. For example, "If Nick is 52 inches tall, then how tall is Matt?" Students should respond with 50 inches. This example will help students understand why the expression $n - 2$ represents the context.



Question 6 is a good place to offer an extension. Ask students how they think they could solve the inequality using algebraic thinking.

14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Component Overview: Students consider constraints on the solution sets of inequalities based on the contexts. Students also determine whether the minimum or maximum values will be included or excluded based on whether the circle is open or closed. Students reason about whether solutions make sense in the context of problems.



14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Inequalities have an infinite number of solutions. The set of all the solutions of an inequality is called the **solution set**.

The **solution set** of an inequality includes the values that, when substituted for the variable, make the inequality true.

One way to represent the solution set of an inequality is by shading the values on the number line, beginning with the minimum or maximum value, and including all the values to the right or left of it that are included in the solution set.

1. Complete each statement.

A minimum or maximum value represented by an open circle means the value is ☐ included in ☒ excluded from the solution set.

A minimum or maximum value represented by a closed circle means the value is ☒ included in ☐ excluded from the solution set.

Graphs representing the solution set of inequalities are helpful when interpreting them in real-world contexts. However, sometimes there are parts of the solution set that may not make sense in a given context.



Before moving, check for understanding regarding the definition of solution set. Ask students what “an infinite number of solutions” means.

14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts (continued)

For example, to ride the Bay Slides at Typhoon Lagoon, guests must be 60 inches or shorter. The graph below represents the graph of $h \leq 60$ in this situation, where h is the height of guests that can ride the Bay Slides.



2. What does 60 represent in this situation?
60 represents the maximum height of guests that can go on Bay Slides.
3. What does the arrow pointing to the left on the number line represent?
The arrow pointing towards left represent the heights of guests that can ride on Bay Slides.
4. Describe at least two values that are part of the solution set of $h \leq 60$ that do not make sense in this context.
Answers will vary. $h = -2$ or $h = -0.25$ because you cannot have negative heights.



Sometimes there are values that are part of the mathematical solution set that do not make sense in the context. In those cases, the variable may have a **constraint** (restriction or limitation) placed on it so that the mathematical model is realistic.

The Exploration demonstrated the inequalities $n - 2 \geq 48$ and $n \geq 50$ are equivalent because all of the same values of n make both inequalities true. Both inequalities were also represented using the same graph.

Inequalities that have the same solution set are called equivalent inequalities.



- What is the minimum height you would expect to see on a water slide?
- What heights are too low?
- Are negative solutions reasonable in the context of this question? Why or why not?



Students may struggle to comprehend the word “constraint.” Consider using a Vocabulary Improvement Strategy to connect the meaning of the word to other contexts and a visual representation of what it means to the student.

14.9.4 Exploration: Properties of Inequality

Component Overview: Students explore how the asymmetric property of inequalities allows them to reverse the order of the inequality. Students also explore how the addition and subtraction properties of inequality allow them to add or subtract the same number to both sides of the inequality.



14.9.4 Exploration: Properties of Inequality

Three inequalities are shown.

$$x > -2.3 \quad -2.3 > x \quad -2.3 < x$$

1. Circle the inequalities that represent the same solution set.
2. Complete the statement.

For both inequalities, all the values of x that are greater than -2.3 make both inequalities true.

These two inequalities are equivalent by the **asymmetric property of inequality**.



The **asymmetric property of inequality** states that if $a > b$, then $b < a$.

3. Use the asymmetric property of inequality to write an inequality equivalent to $q + \frac{2}{5} \leq \frac{7}{10}$.

$$\frac{7}{10} \geq q + \frac{2}{5}$$



Consider having students practice reading inequalities aloud from right to left and left to right.



The Exploration is intentionally scaffolded to make connections between solving equations and inequalities. The Exploration also is scaffolded to introduce the properties of inequality that students apply when solving addition and subtraction inequalities. While circulating, support students who may need help.

14.9.4 Exploration: Properties of Inequality (continued)

4. Four true inequalities are given in the table. Work with your partner to complete the table by following the steps indicated for each inequality and then answering the questions.

Given Inequality	Step	What is the new inequality?	Is the new inequality true?
$-6 < -2$	Add 8.1 to both sides	$2.1 < 6.1$	yes
$-5 < \frac{3}{4}$	Subtract 2 from both sides	$-7 < -1\frac{1}{4}$	yes
$7.45 > 7$	Add $3\frac{1}{5}$ to both sides	$10.65 > 10\frac{1}{5}$	yes
$0 > -\frac{2}{3}$	Subtract $\frac{5}{6}$ from both sides	$-\frac{5}{6} > -1\frac{1}{2}$	yes



Students should conclude that adding or subtracting the same value to both sides of an inequality results in a true statement. Provide a visual representation of why this is true by moving the two values in the inequality on the number line the same distance.

5. Use your work on the previous problem to complete the description of each property.

Property of Inequality	Description
Addition property of inequality	When the same value is added to both sides of a true inequality, ... <i>the new inequality is also true.</i>
Subtraction property of inequality	When the same value is subtracted from both sides of a true inequality, ... <i>the new inequality is also true.</i>

14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction

Component Overview: Students begin by completing a table for a given expression. Using the table and the relationship of the expression to the number -2 , the boundary point, solutions, and graph of the relationship is determined. Then students learn how to solve inequalities using the addition and subtraction properties of inequality.



14.9.5 Guided Instruction: Solving One-Step Inequalities with Addition and Subtraction

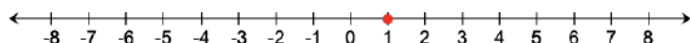
- Complete the table to determine the value of $x - 3$ for each value of x .

x	-4	-3.2	-2	$-\frac{1}{2}$	0	1	2	$3\frac{6}{7}$	4
$x - 3$	-7	-6.2	-5	$-3\frac{1}{2}$	-3	-2	-1	$\frac{6}{7}$	1

- For which value of x is the equation $x - 3 = -2$ true?

1

- Represent the solution to the equation $x - 3 = -2$ on the number line.



- For which values of x in the table is the inequality $x - 3 > -2$ true?

2; $3\frac{6}{7}$; 4



Connect the answer to question 2 to the table of values.



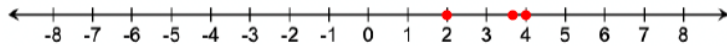
Compare and Connect

Highlight the connection between the equation and the inequality by considering the following questions.

- What changes when the equality symbol changes to an inequality symbol?
- How do the graphs of the solutions compare?

14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction (continued)

5. Represent the values from the table that are part of the solution set of $x - 3 > -2$ on the number line.

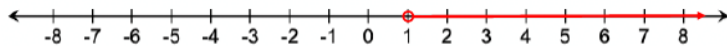


The solution set of $x - 3 > -2$ includes more values than shown on the previous number line.



Highlight the difference between the graph of the solution to an equation vs an inequality. Students do not often graph solutions to an equation on a number line. However, seeing the difference will provide a visual entry point.

6. Graph the solution set for the inequality.



7. Write a value on the line to show an equivalent inequality represented by the graph above.

$$x > \underline{1}$$



Students may answer $x > 2$ since that is the place where the value of $x - 3$ first surpasses the number -2 . Remind students that there are values between 1 and 2. Ask whether $1.1 - 3 > -2$.

The solution set of the inequality $x - 3 > -2$ is all the values of x that are greater than 1, or $x > 1$.

14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction (continued)

This solution set can be determined algebraically by adding the same value to both sides of the inequality.

8. What value can be added to both sides of the inequality $x - 3 > -2$ to isolate x ?

3

Adding the same value to both sides of an inequality can be used to solve an inequality by the **addition property of inequality**.



Addition Property of Inequality

If $a > b$, then $a + c > b + c$.

The **addition property of inequality** states that if the same value is added to both sides of an inequality, the result will be an equivalent inequality that is also true.

The **subtraction property of inequality** can also be used to solve an inequality.



Subtraction Property of Inequality

If $a > b$, then $a - c > b - c$.

The **subtraction property of inequality** states that if the same value is subtracted from both sides of an inequality, the result will be an **equivalent** inequality that is also **true**.

These properties can be used to solve one-step inequalities algebraically.



Students may benefit from first using concrete values instead of abstract variables. Consider having students substitute different integers for a , b , and c for a better understanding of addition and subtraction properties of inequality.



To expand students' thinking, have them discuss what other properties closely relate to these two properties of inequality. Students should be able to relate the following properties.

- The addition property of equality and the addition property of inequality.
- The subtraction property of equality and the subtraction property of inequality.

Prompt students to anticipate any other properties previously learned that might relate to inequalities.

14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction (continued)

9. Use the inequality $-12.25 > y - 2.4$ to complete the following.

- a. Find the solution set by isolating the variable. Show your work.

$$-12.25 > y - 2.4$$

$$\underline{+2.4} \quad \underline{+2.4}$$

$$-9.85 > y + 0$$

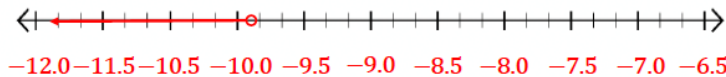
$$-9.85 > y$$

$$y < -9.85$$



Now that students have been formally introduced to the addition and subtraction properties of inequality, they should use them when solving inequalities. Ask students to compare the efficiency of algebraic methods compared to using substitution and Guess and Test.

- b. Graph the solution set on the number line.



- c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

Answers will vary.

$$-12.25 > -10 - 2.4$$

$$-12.25 > -12.4$$

$$-12.25 > -12 - 2.4$$

$$-12.25 > -14.4$$



Highlight the importance of checking answers by testing values from the graph in the inequality.

14.9.6 Wrap-Up: Cool Down

Component Overview: Students reflect on their mastery of learning targets by completing a Know, Wonder, and Learned (K-W-L chart) table. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



14.9.6 Wrap-Up: Cool Down

- 1. Complete the table.
Answers will vary.

The terms and information used in today's lesson that I knew were...	Something I wondered during today's lesson was...	Something I learned during today's lesson was...



For students who struggle to represent their conceptual understanding through writing, consider providing alternative ways to assess learning for this lesson. Students could use an audio recorder or speech-to-text software, draw and label a diagram, or provide examples. All forms show an equal level of rigor and conceptual understanding.